11.4/ The Comparison Tests

This test can be used just for a series of positive terms

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Example

- **Test the following series for convergence or divergence**

$$\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$$

Geometric-series

$$\sum_{n=1}^{\infty} ar^{n-1}$$

↓ conv. if $|r| < 1$

and div. if $|r| \geq 1$

$$S = \frac{\text{First term}}{1 - r}$$

p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

↓ conv. if $|p| > 1$

and div if $|p| \leq 1$

$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ geo.

$\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^2$ p-series

$$\sum_{n=1}^{\infty} \frac{1}{2^n + 1} \quad ?$$

$$\overset{①}{<} \sum_{n=1} \frac{1}{2^n}$$

② conv. geometric series

⇒ By the comparsion test our series

$\sum \frac{1}{2^n + 1}$ is **conv.**

$\sum \left(\frac{1}{2}\right)^n$

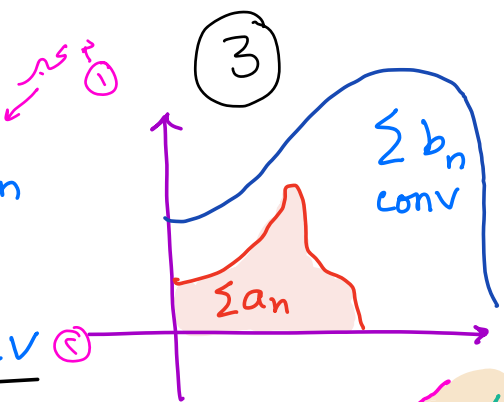
↳ geometric

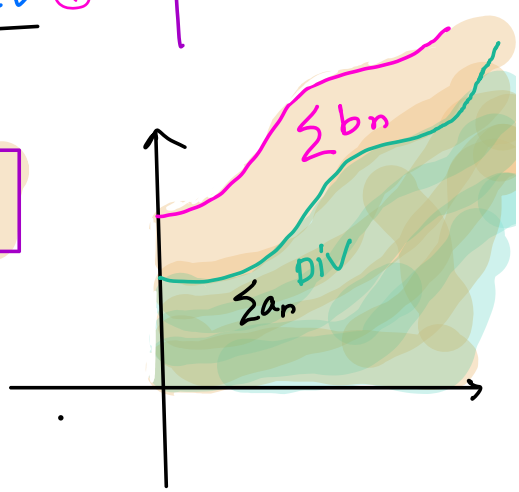
$\sum \frac{1}{n^2} = \sum \left(\frac{1}{n}\right)^2$

↳ p-series

Remark: comparsion test

1. your series $\sum a_n \leq \sum b_n$ (②)
   - if conv ⑤ → $\sum a_n$ is conv.
   - if div ⑤ ← proof ① $\sum a_n \leq$ your series $\sum b_n$ ② → $\sum b_n$ is div.





The Comparison Tests

The Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum a_n$ is also divergent.

Remark:

We can not use this test For the series:

$$\rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{1}{n} = -\frac{1}{1} + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} \cdots$$

because it is contain some negative terms

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comparsion test:

$\sum_{n=1}^{\infty} a_n$

$\sum_{n=1}^{\infty} b_n$ [geometric, p-series]

greater than my series in the question

less than my series

+ conv.

$\Rightarrow \sum a_n$ conv.

+ div.

$\sum b_n$ div

Ex: Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$ For convergent

Solution:

$$\sum_{n=1}^{\infty} \frac{1}{2^n + 1} < \sum_{n=1}^{\infty} \frac{1}{2^n}$$

bigger ①

geom. serie

+

$|r| = \frac{1}{2} < 1$

↳ conv.

$\Rightarrow$ our series which is $\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$ is conv.

$$\boxed{\frac{1}{3}} > \boxed{\frac{1}{5}}$$

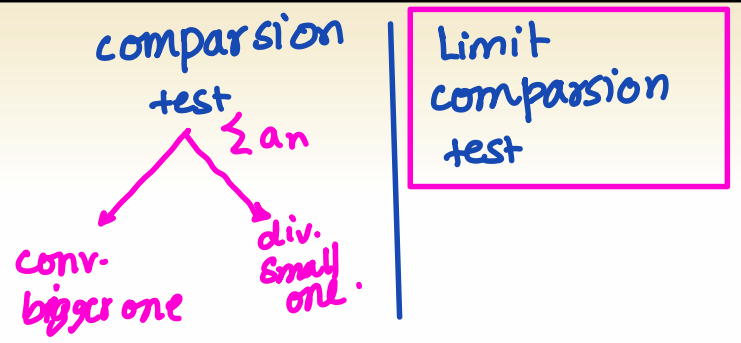
The Comparison Tests

- In using the Comparison Test we must, of course, have some known series $\Sigma\, b_n$ for the purpose of comparison. Most of the time we use one of these series:

- A *p*-series [$\Sigma\ 1/n^p$ converges if $p > 1$ and diverges if $p \leq 1$]
  A geometric series [$\Sigma\ ar^{n-1}$ converges if $|r| < 1$ and
  diverges if $|r| \geq 1$]

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Remark :



1. **The terms of the series being tested must be smaller than those of a convergent series or larger than those of a divergent series.**

2. **If the terms are larger than the terms of a convergent series or smaller than those of a divergent series, then the Comparison Test doesn't apply.**

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Remark: Note that:

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1} \quad ? \quad > \quad \sum_{n=1}^{\infty} \frac{1}{2^n}$$

geom. ↓ conv. series

You can not use the comparsion test

Remark:

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- Consider, the series
- $$\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$$

The inequality $\frac{1}{2^n - 1} > \frac{1}{2^n}$

is useless as far as the Comparison Test is concerned because $\Sigma\, b_n = \Sigma \left(\frac{1}{2}\right)^n$ is convergent and $a_n > b_n$.

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The comparsion test

If you have $\sum a_n$, $\sum b_n$

$b_n \geq a_n > 0 \quad \forall n$

(big) (small)

then

① If $\sum b_n$ conv. $\rightarrow \sum a_n$ conv.

② if $\sum a_n$ div. $\rightarrow \sum b_n$ div.

③ if $\sum b_n$ is div $\not\rightarrow \sum a_n$ div

④ if $\sum a_n$ is con $\not\rightarrow \sum b_n$ con

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Example 1

Ex:

- Determine whether the series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$$

?

① bigger

$$\longleftrightarrow \sum_{n=1}^{\infty} \frac{5}{2n^2}$$

$$= \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

p-series with $p = 2$

② so, it is conv.

$\Rightarrow$ our series $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ is conv.

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Note:

- Although the condition $a_n \le b_n$ or $a_n \ge b_n$ in the Comparison Test is given for all n, we need verify only that it holds for $n \ge N$, where N is some fixed integer, because the convergence of a series is not affected by a finite number of terms.

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The Limit Comparison Test

The Limit Comparison Test Suppose that $\Sigma\, a_n$ and $\Sigma\, b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

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Ex: Determine the convergence or divergence for the following series

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1} \quad , \quad \sum_{n=1}^{\infty} \frac{1}{2^n}$$

Sol:

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1} \quad , \quad \sum_{n=1}^{\infty} \frac{1}{2^n}$$ → conv. geometric series

using the limit comparsion test we have:

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{2^n - 1}}{\frac{1}{2^n}}$$

$$= \lim_{n\to\infty} \frac{1}{2^n - 1} \cdot \frac{2^n}{1} = \lim_{n\to\infty} \frac{2^n}{2^n - 1} = \frac{\infty}{\infty}$$

indeterminat Form

$$= \lim_{n\to\infty} \frac{\frac{2^n}{2^n}}{\frac{2^n}{2^n} - \frac{1}{2^n}} = \lim_{n\to\infty} \frac{1}{1 - \frac{1}{2^n}}$$

$$= \frac{1}{1 - \lim_{n\to\infty} \frac{1}{2^n}} = \frac{1}{1 - 0} = 1 > 0$$

By limit comparsion test both series are converg. or div. as $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is conv $\Rightarrow \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

Example 3

- Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence or divergence.

- Solution: $a_n = \frac{1}{2^n - 1}$ $b_n = \frac{1}{2^n}$
- We use the Limit Comparison Test with

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{1/(2^n - 1)}{1/2^n}$$

$$= \lim_{n\to\infty} \frac{2^n}{2^n - 1}$$

- and obtain

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Ex: Determine whether the series

$$\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}} = \sum_{n=1}^{\infty} a_n$$

is convergent or divergent

Solution:

$$\sum_{n=1}^{\infty} \frac{2n^2}{n^{5/2}} = \sum_{n=1}^{\infty} \frac{2}{n^{5/2-2}}$$

$$= \sum_{n=1}^{\infty} \frac{2}{n^{1/2}} = 2\sum_{n=1}^{\infty} \frac{1}{n^{1/2}} = \sum_{n=1}^{\infty} b_n$$

p-series with $p = 1/2 < 1$

so it is divergent.

using limit comparsion test we have:

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{2}{n^{1/2}}}$$

$$= \lim_{n\to\infty} \frac{2n^2+3n}{\sqrt{5+n^5}} \cdot \frac{n^{1/2}}{2}$$

$$= \lim_{n\to\infty} \frac{n^{5/2}+3n^{3/2}}{2\sqrt{5+n^5}} = \frac{\infty}{\infty}$$

$$= \lim_{n\to\infty} \frac{\frac{2n^{5/2}}{n^{5/2}} + \frac{3n^{3/2}}{n^{5/2}}}{2\sqrt{\frac{5}{n^5}+\frac{n^5}{n^5}}}$$

$$= \lim_{n\to\infty} \frac{2+\frac{3}{n}}{2\sqrt{\frac{5}{n^5}+1}}$$

$$= \frac{2+0}{2\sqrt{0+1}} = \frac{2}{2} = 1 > 0$$

using limit comparsion test both series are conv. or both are div.

and as $\sum_{n=1}^{\infty} \frac{2}{n^{1/2}}$ is div.

$$\Rightarrow \sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}}$$ is div.

Example 3 – *Solution*

- $$= \lim_{n \to \infty} \frac{1}{1 - 1/2^n}$$

$$= 1 > 0$$

- Since this limit exists and $\Sigma\ 1/2^n$ is a convergent geometric series, the given series converges by the Limit Comparison Test.

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Remark:

1. comparsion test

$$\sum a_n \quad < \quad \sum b_n$$
$$\hookrightarrow \text{conv.}$$

$$\Rightarrow \sum a_n \text{ conv.}$$

limit comparsion test.

$$\sum a_n \quad > \quad \sum b_n$$
$$\hookrightarrow \text{div}$$

$$\Rightarrow \sum a_n \text{ div.}$$

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2. limit comp. test

$$\sum a_n \ , \ \sum b_n$$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$$

$\Rightarrow \sum a_n$ conv. if $\sum b_n$ is conv

$\Rightarrow \sum a_n$ div. if $\sum a_n$ is div.
